Solving Equations: Distributive Property & Variables on Both Sides — Answer Key

Type 1 Distribute First

1. Distribute: $2x + 6 = 14$. Undo $+6$: $2x = 8$. Divide by 2.
Answer: $x = 4$

2. Distribute: $3x - 6 = 12$. Undo -6: $3x = 18$. Divide by 3.
Answer: $x = 6$

3. Distribute: $4x + 4 = 20$. Undo $+4$: $4x = 16$. Divide by 4.
Answer: $x = 4$

4. Distribute: $5x - 15 = 10$. Undo -15: $5x = 25$. Divide by 5.
Answer: $x = 5$

5. Distribute: $2x + 10 = 18$. Undo $+10$: $2x = 8$. Divide by 2.
Answer: $x = 4$

6. Distribute: $3x + 12 = 27$. Undo $+12$: $3x = 15$. Divide by 3.
Answer: $x = 5$

7. Distribute: $6x - 6 = 24$. Undo -6: $6x = 30$. Divide by 6.
Answer: $x = 5$

8. Distribute: $4x - 20 = 8$. Undo -20: $4x = 28$. Divide by 4.
Answer: $x = 7$

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Type 2 Variables on Both Sides

9. Subtract $3x$ from both sides: $2x + 2 = 10$. Then $2x = 8$.
Answer: $x = 4$

10. Subtract $4x$ from both sides: $3x - 3 = 12$. Then $3x = 15$.
Answer: $x = 5$

11. Subtract $2x$ from both sides: $4x + 1 = 13$. Then $4x = 12$.
Answer: $x = 3$

12. Subtract $5x$ from both sides: $4x - 4 = 20$. Then $4x = 24$.
Answer: $x = 6$

13. Subtract x from both sides: $3x + 7 = 16$. Then $3x = 9$.
Answer: $x = 3$

14. Subtract $3x$ from both sides: $5x = 35$. Then $5x = 35$.
Answer: $x = 7$

15. Subtract $2x$ from both sides: $15 = 3x$. Then $15 = 3x$.
Answer: $x = 5$

16. Subtract $7x$ from both sides: $3x - 6 = 9$. Then $3x = 15$.
Answer: $x = 5$

Type 3 Distribute, Then Variables on Both Sides

17. Distribute: $2x + 8 = x + 11$. Subtract x: $x + 8 = 11$, so $1x = 3$.
Answer: $x = 3$

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18. Distribute: $3x - 3 = x + 7$. Subtract x: $2x - 3 = 7$, so $2x = 10$.
Answer: $x = 5$

19. Distribute: $4x + 8 = 2x + 14$. Subtract $2x$: $2x + 8 = 14$, so $2x = 6$.
Answer: $x = 3$

20. Distribute: $5x - 10 = 3x + 4$. Subtract $3x$: $2x - 10 = 4$, so $2x = 14$.
Answer: $x = 7$

21. Distribute: $2x + 12 = 4x + 2$. Subtract $2x$: $12 = 2x + 2$, so $10 = 2x$.
Answer: $x = 5$

22. Distribute: $3x + 9 = 5x + 1$. Subtract $3x$: $9 = 2x + 1$, so $8 = 2x$.
Answer: $x = 4$

23. Distribute: $6x - 6 = 2x + 10$. Subtract $2x$: $4x - 6 = 10$, so $4x = 16$.
Answer: $x = 4$

24. Distribute: $4x - 12 = x + 3$. Subtract x: $3x - 12 = 3$, so $3x = 15$.
Answer: $x = 5$

Type 4 Distribute, Then Combine Like Terms

25. Distribute: $2x + 6 + x = 21$. Combine: $3x + 6 = 21$, so $3x = 15$.
Answer: $x = 5$

26. Distribute: $3x - 6 + 2x = 14$. Combine: $5x - 6 = 14$, so $5x = 20$.
Answer: $x = 4$

27. Distribute: $4x + 4 + x = 29$. Combine: $5x + 4 = 29$, so $5x = 25$.
Answer: $x = 5$

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28. Distribute: $2x - 8 + 3x = 17$. Combine: $5x - 8 = 17$, so $5x = 25$.

Answer: $x = 5$

29. Distribute: $5x + 10 - 2x = 25$. Combine: $3x + 10 = 25$, so $3x = 15$.

Answer: $x = 5$

30. Distribute: $3x + 15 + x = 31$. Combine: $4x + 15 = 31$, so $4x = 16$.

Answer: $x = 4$

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